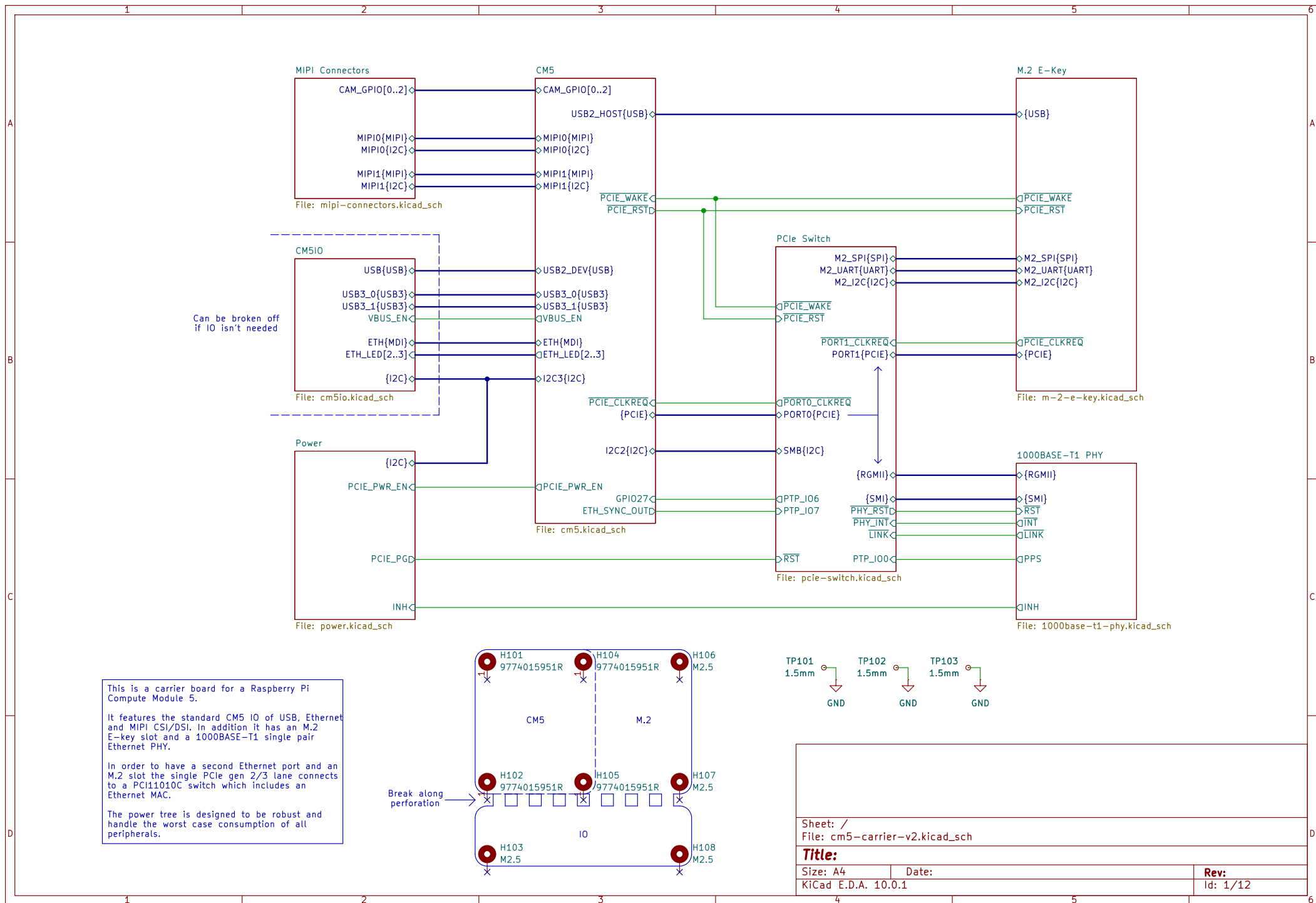
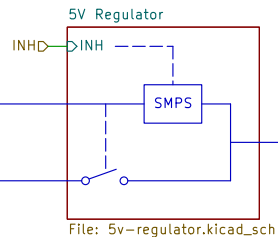
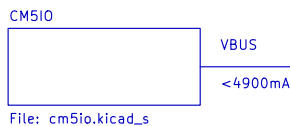
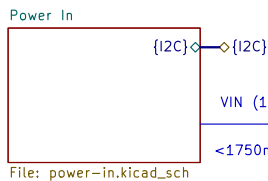




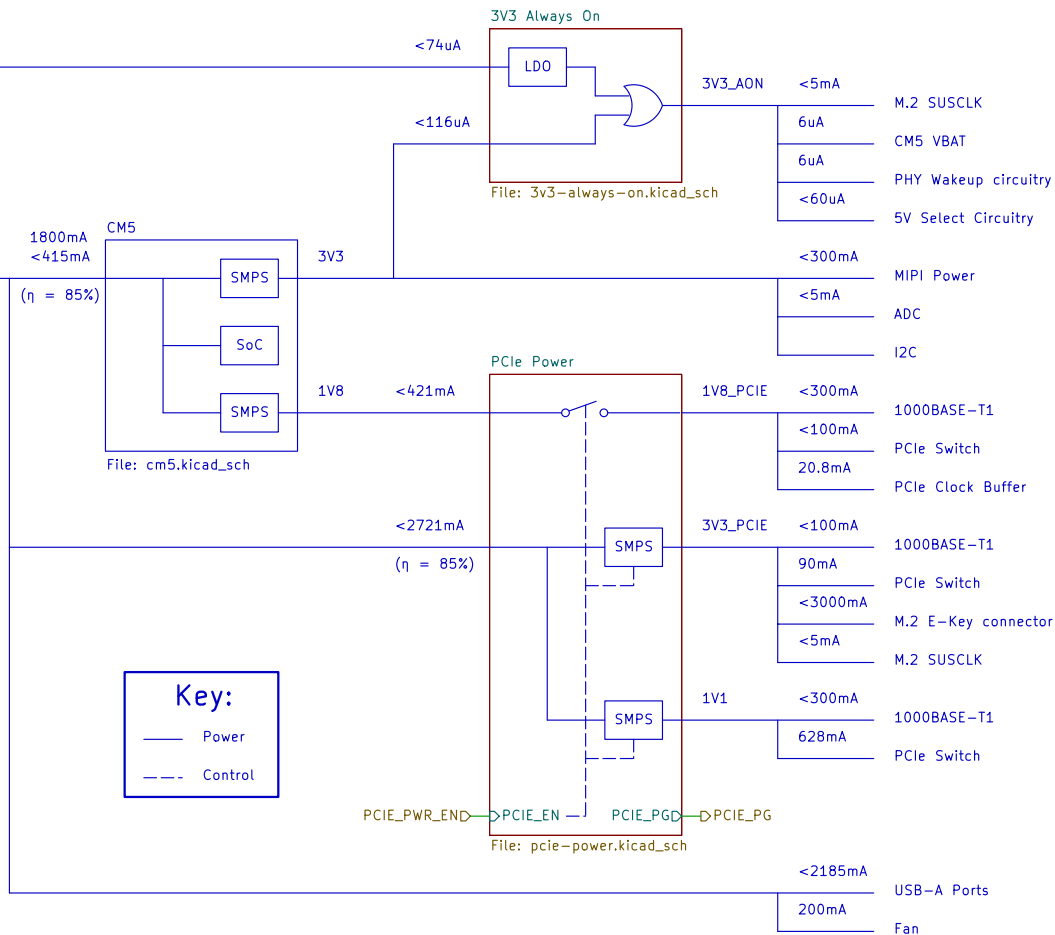
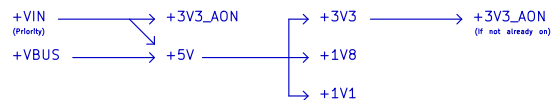


The maximum power that can be consumed by the system is 13W, and the maximum power that could be consumed by peripherals (M.2 & USB-A ports) is 20.4W. Not all sources are able to provide this power so the peripheral power must be limited:

- 5V USB (3A capable): 2W for peripherals
- 5V USB (5A capable): 12W for peripherals
- 12V VIN: 5W for peripherals
- 24V VIN: 20W for peripherals

Note that the USB-A ports cannot consume more than 10W combined.

Power up sequence:



Sheet: /Power/
File: power.kicad_sch

Title:

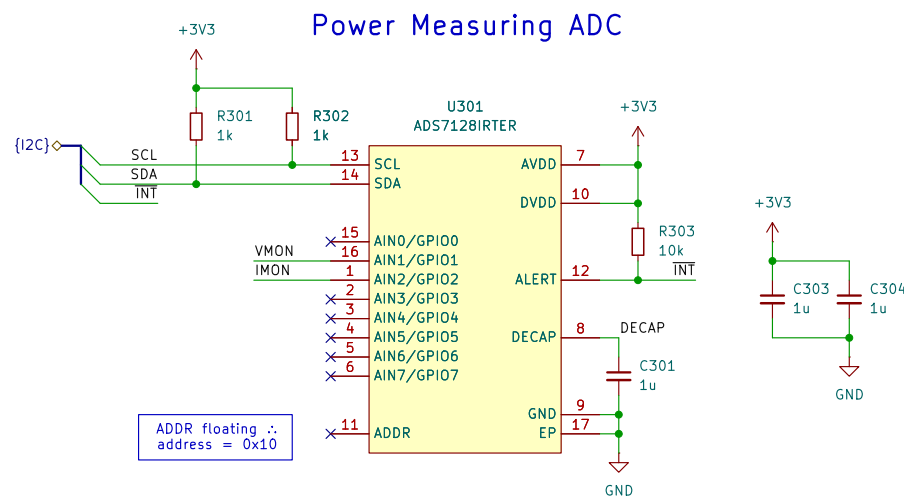
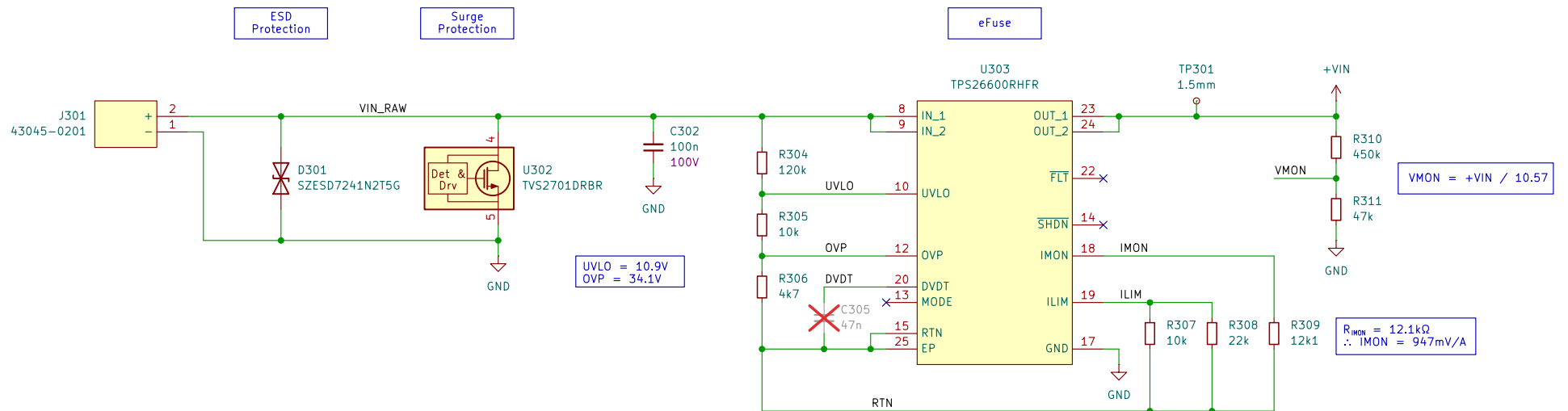
Size: A4

Date:

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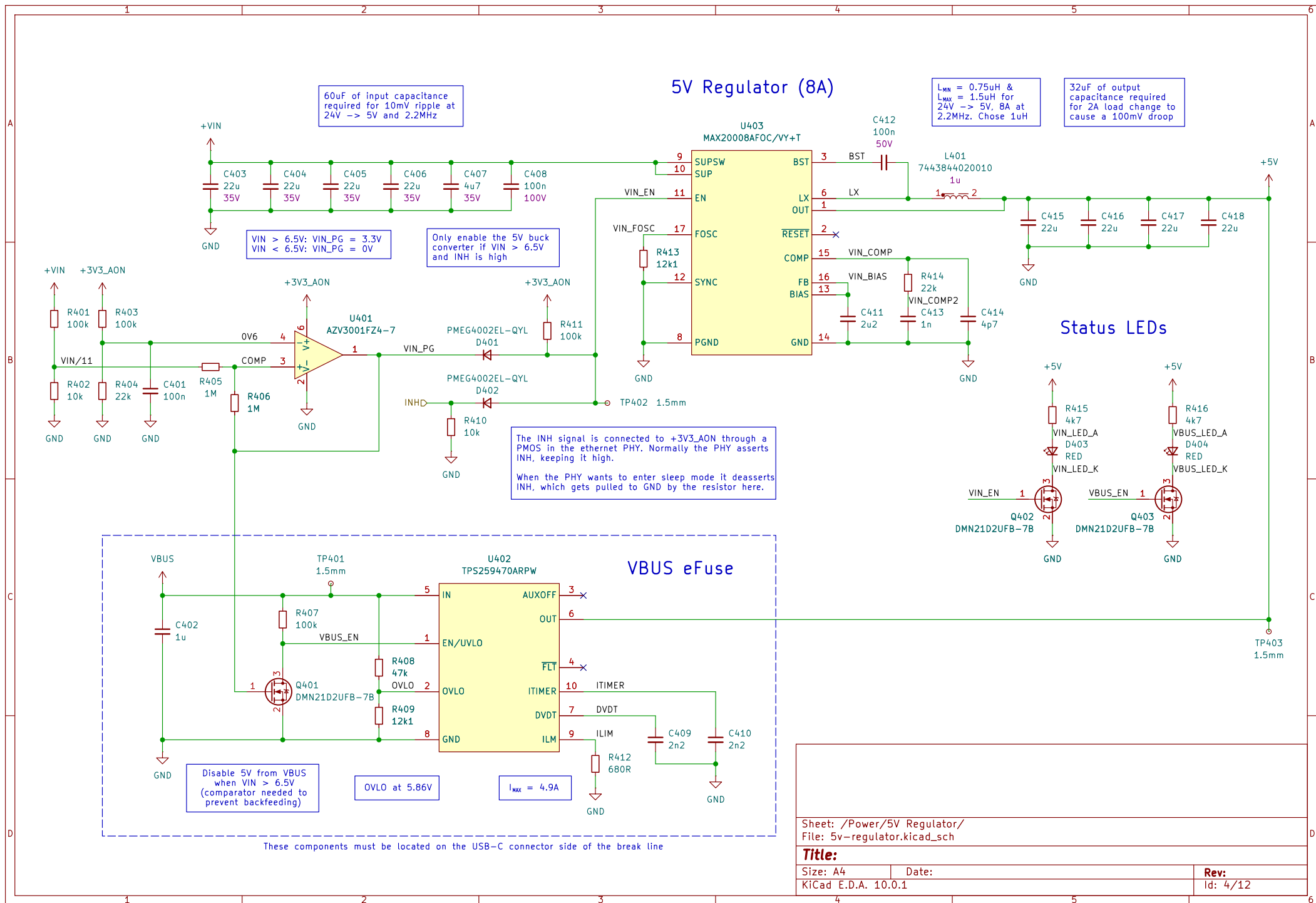
Sheet: /Power/Power In/
File: power-in.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.1

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Id: 3/12



Sheet: /Power/5V Regulator/
File: 5v-regulator.kicad_sch

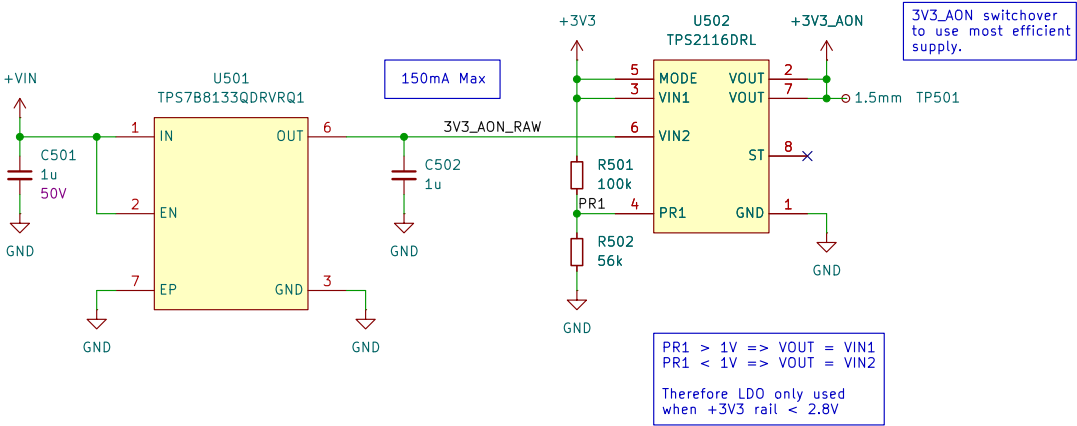
Title:

Size: A4
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Date:

Rev:
Id: 4/12

3V3 Always on supply



Sheet: /Power/3V3 Always On/
File: 3v3-always-on.kicad_sch

Title:

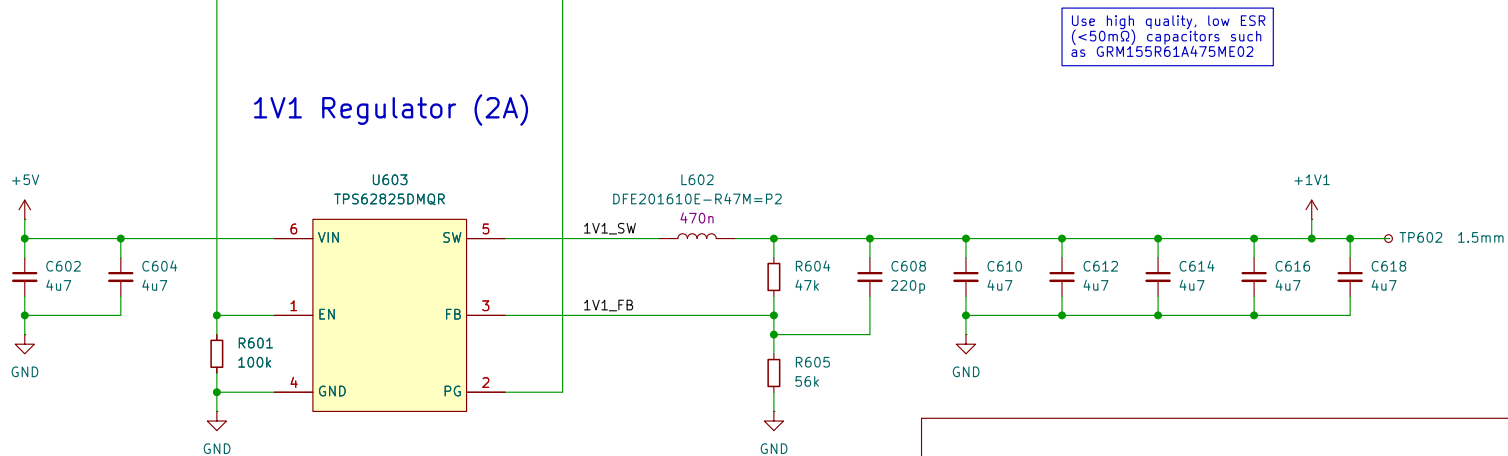
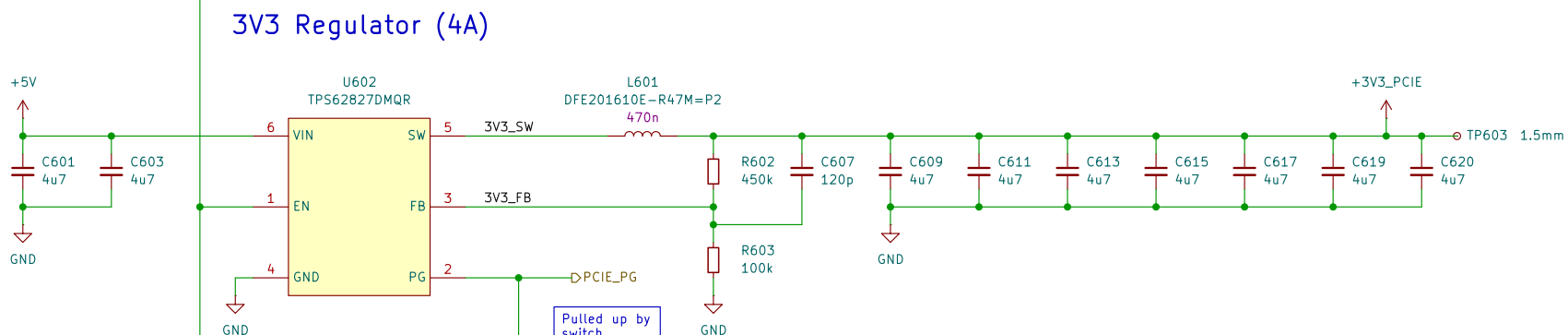
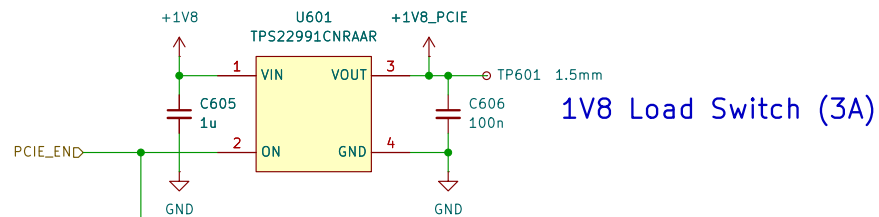
Size: A4

Date:

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Sheet: /Power/PCIE Power/
File: pcie-power.kicad_sch

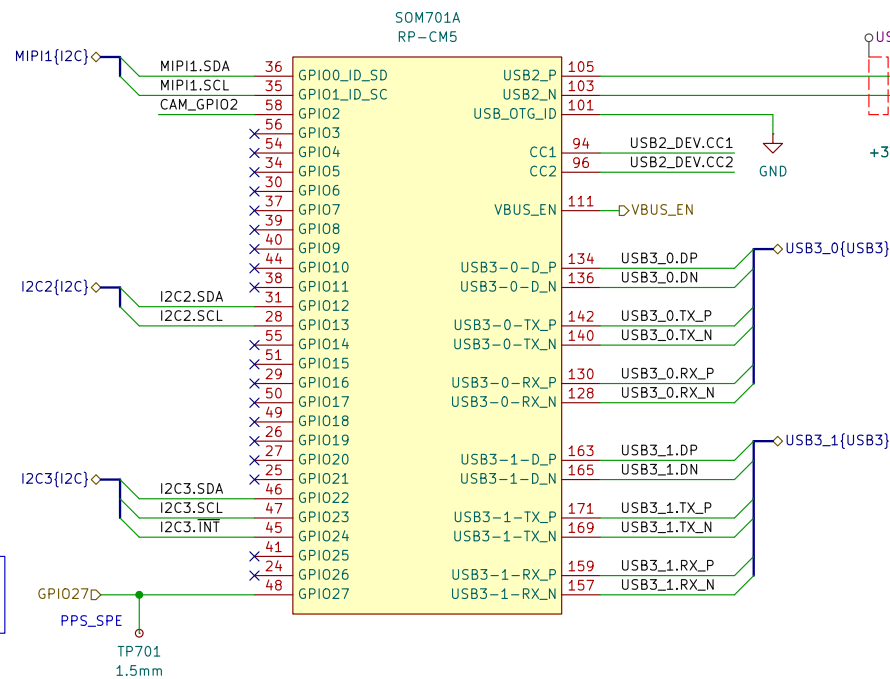
Title:

Size: A4 Date:

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Rev:
Id: 6/12

USB & GPIO

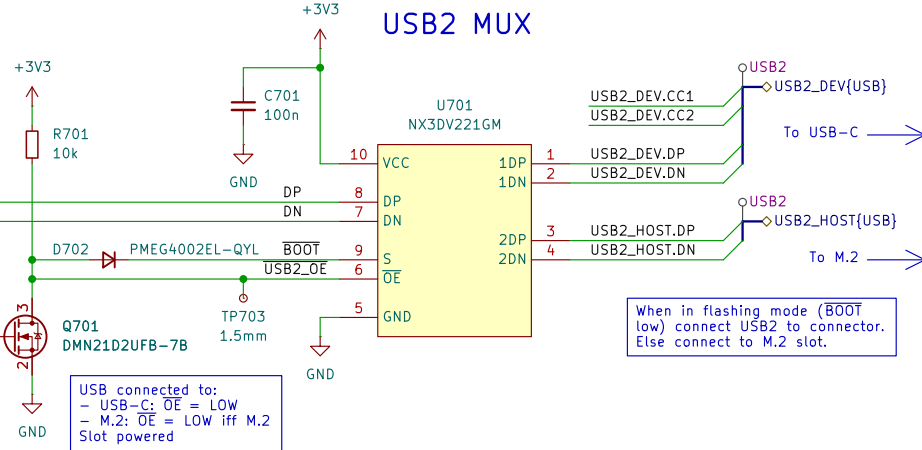


GPIO27 Connects to the PCI11010C's PTP_IO6. PTP_IO6 can be configured to be a PPS output and GPIO27 as a PPS input.

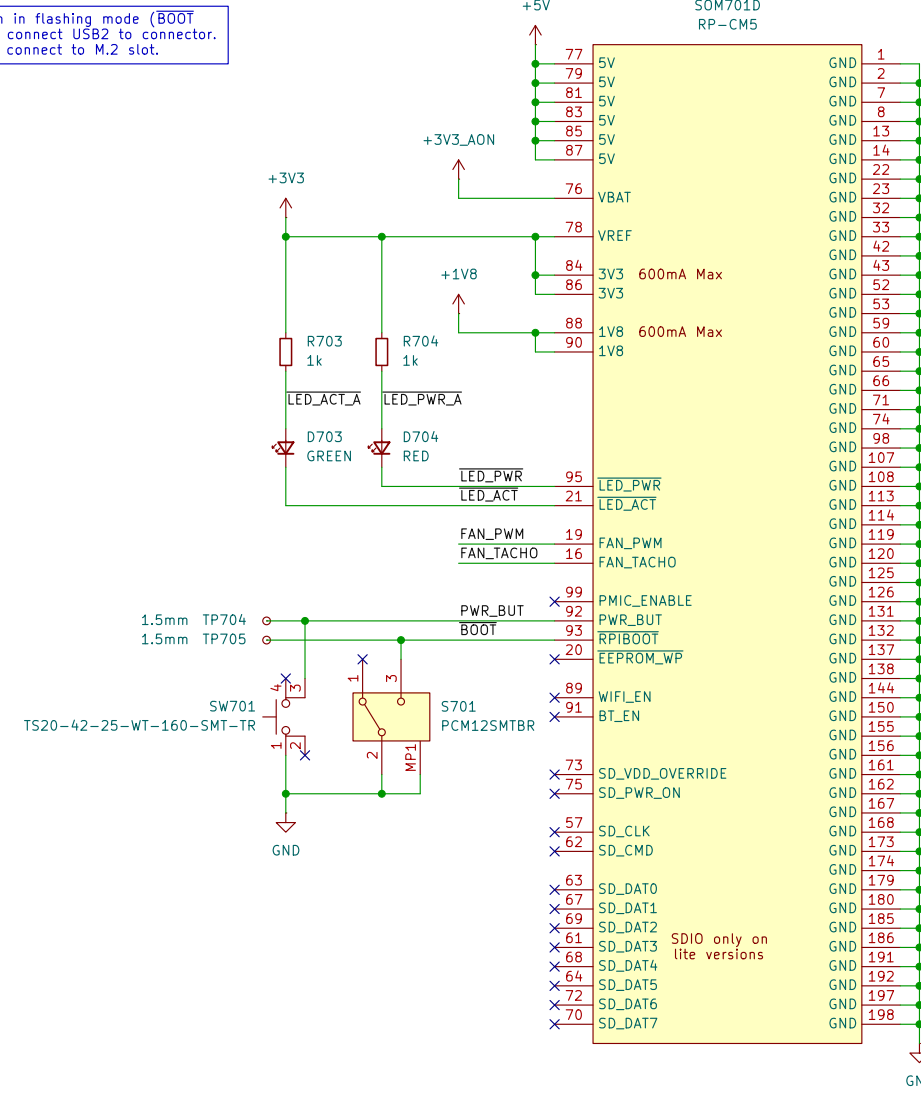
CLKREQ and WAKE are internally pulled up

CM5 PCIe can be either gen 2 (100Ω) or gen 3 (85Ω) so route at 90Ω as recommended by Raspberry Pi.

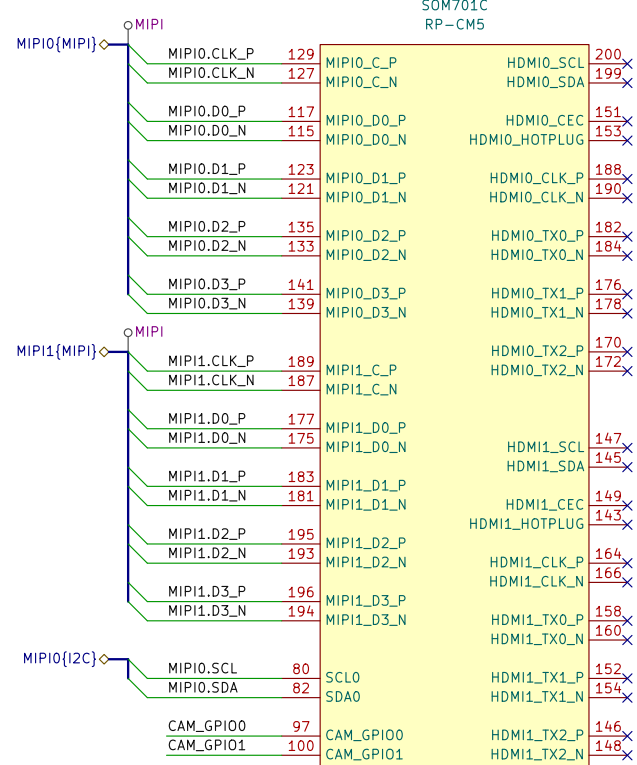
USB2 MUX



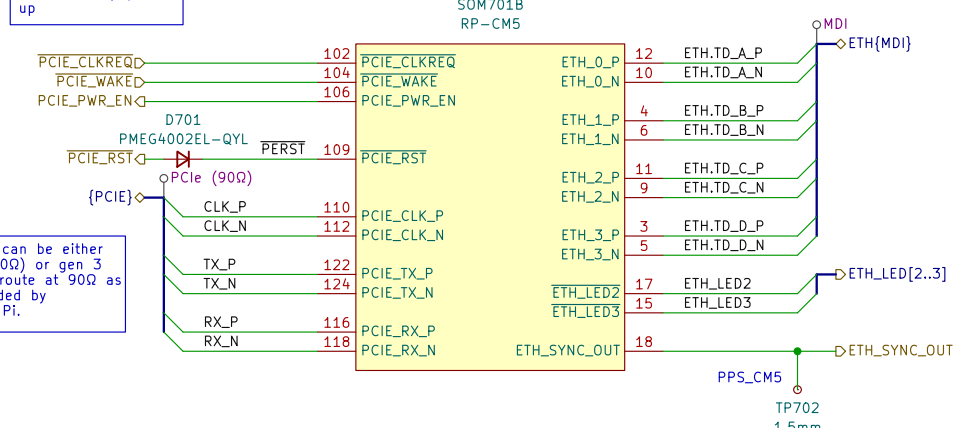
Power & Control



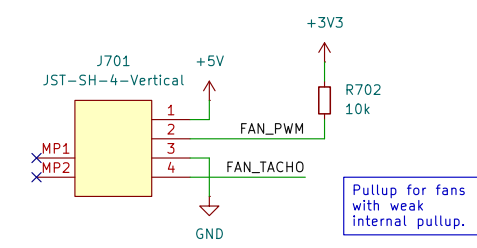
CSI/DSI/HDMI



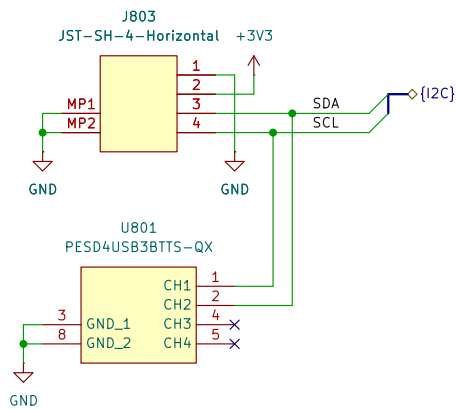
Ethernet & PCIe



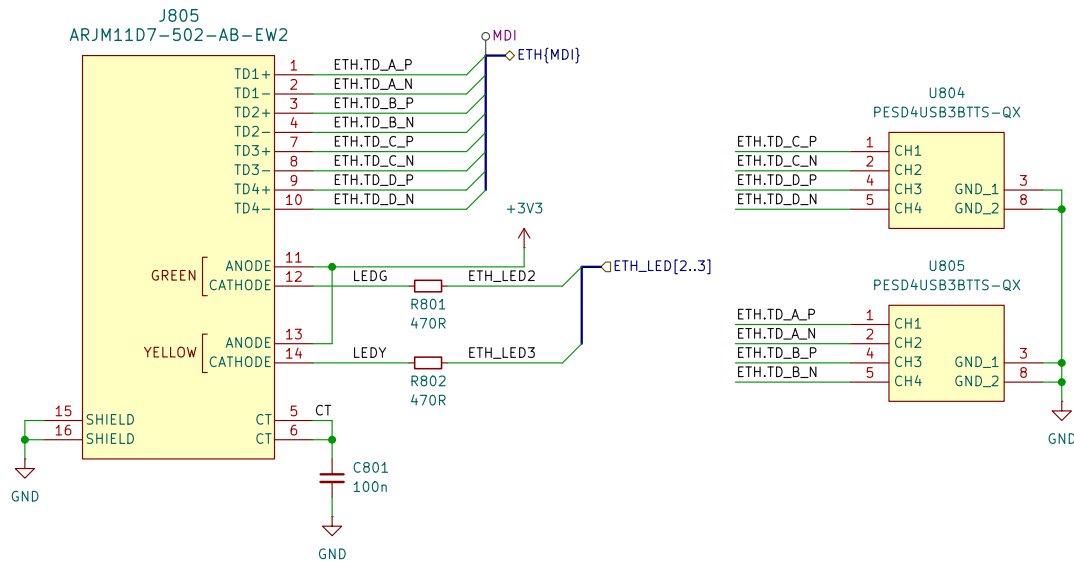
Fan Connector



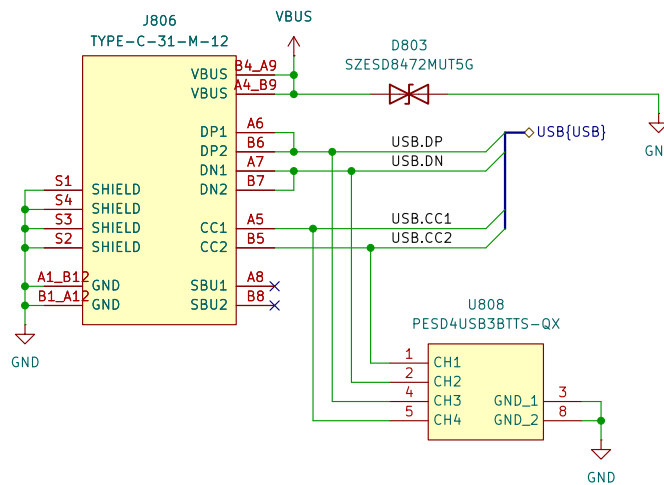
I2C



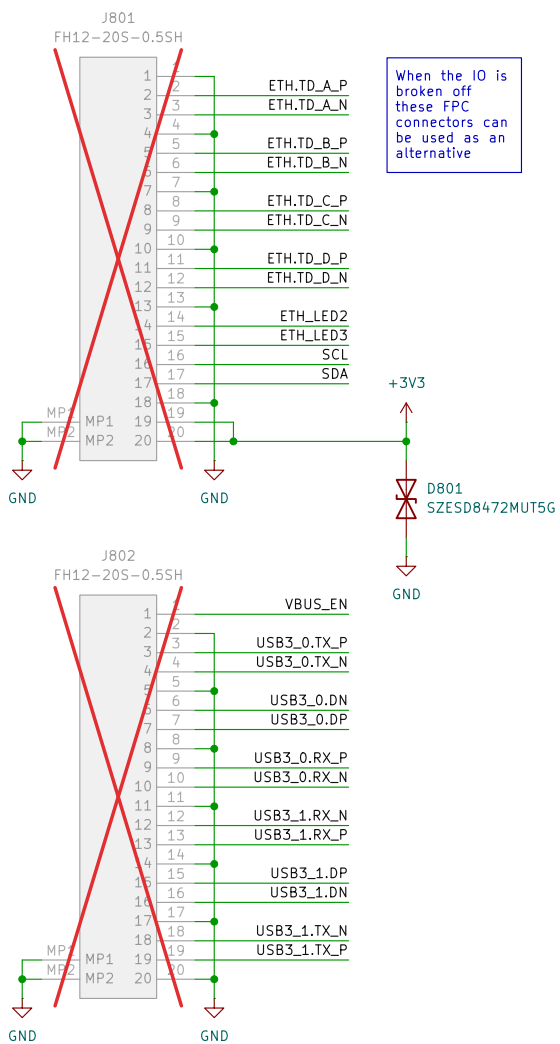
Ethernet RJ45



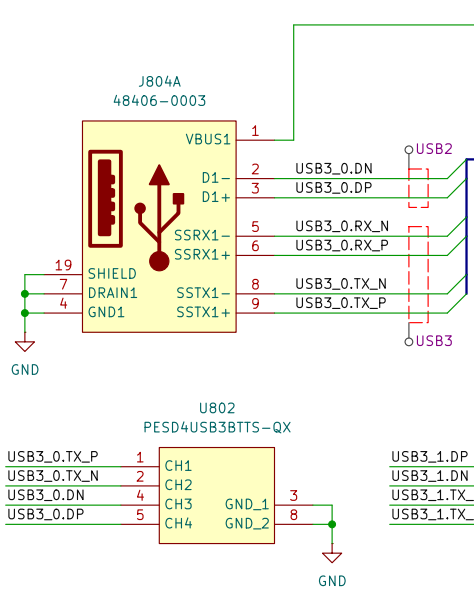
USB2



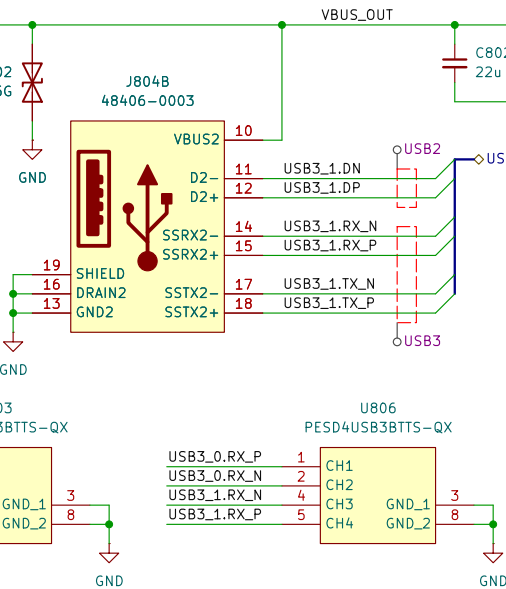
Backup FPC Connectors



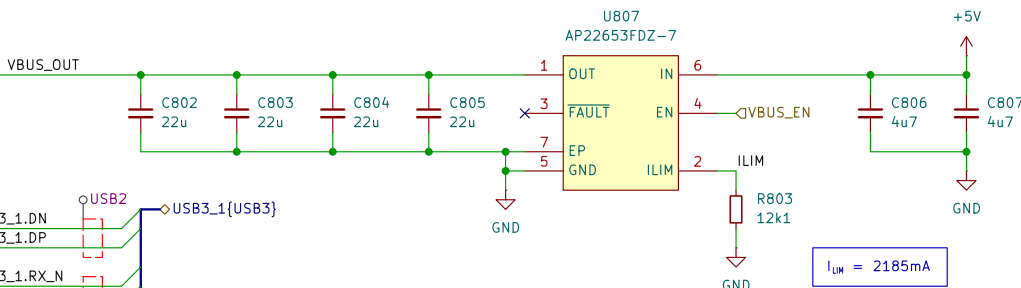
USB3 0

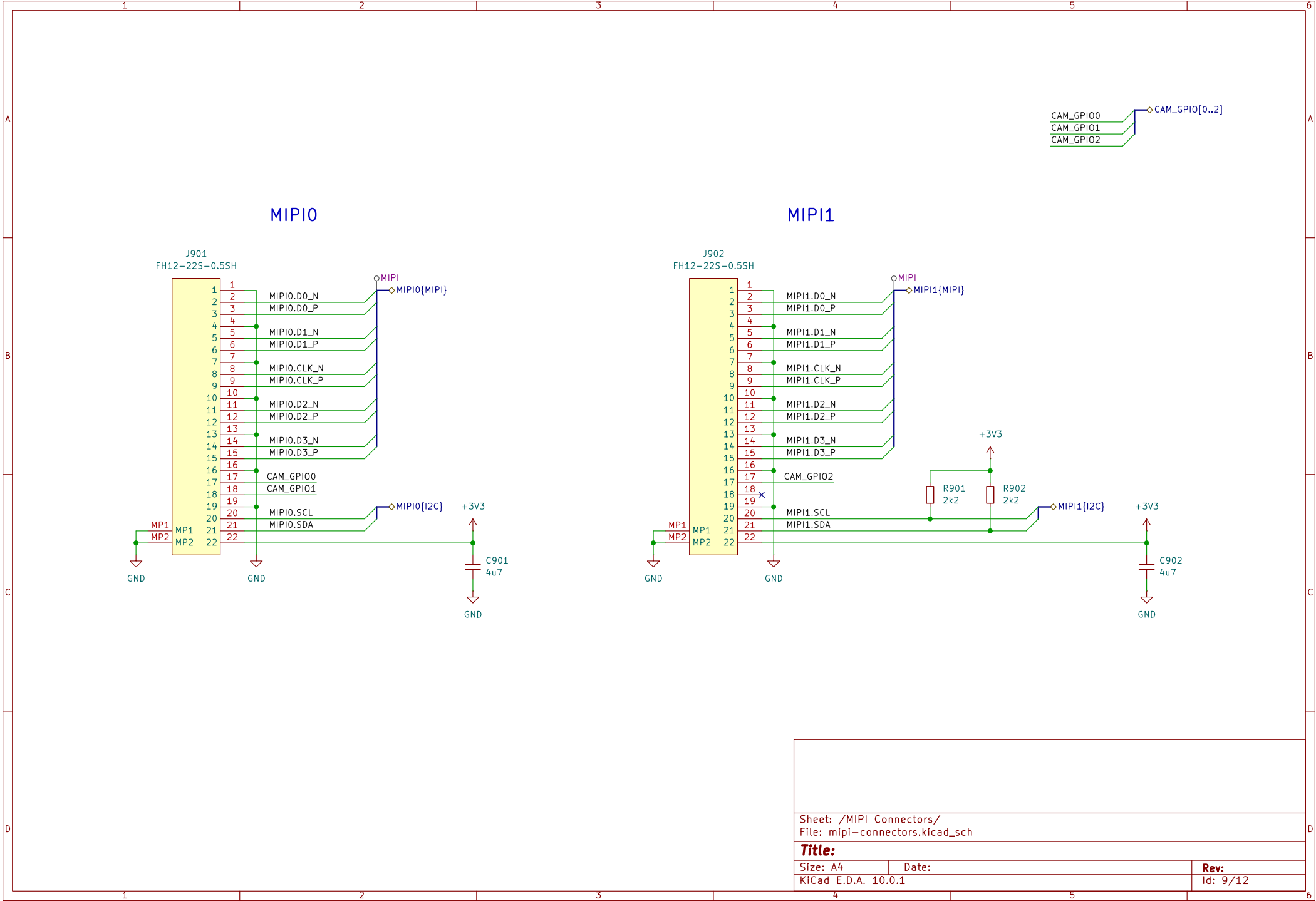


USB3 1



VBUS Load Switch





Sheet: /MIPI Connectors/
File: mipi-connectors.kicad_sch

Title:

Size: A4

Date:

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PCIe CLK termination should be done by the host, however Raspberry Pi / Broadcom doesn't give any information about the driver or termination. Anecdotally it seems to be LP HCSL meaning no extra termination is needed.

OE pins have internal 100k pull down resistors. Diode used to level shift from 3.3V to 1.8V (to save power)

Clock Buffer

P/N swapping to ease routing

The output from the buffer is LP HSCL which is internally terminated meaning no external termination components are required.

PCIe Switch & Ethernet MAC

Pull up straps means the SMBUS interface can be used for configuration on the fly

EEPROM_STRAP Pullup means the PCI11010 will attempt to load config from EEPROM

8KB EEPROM

Sheet: /PCIe Switch/
File: pcie-switch.kicad_sch

Title:

Size: A3

Date:

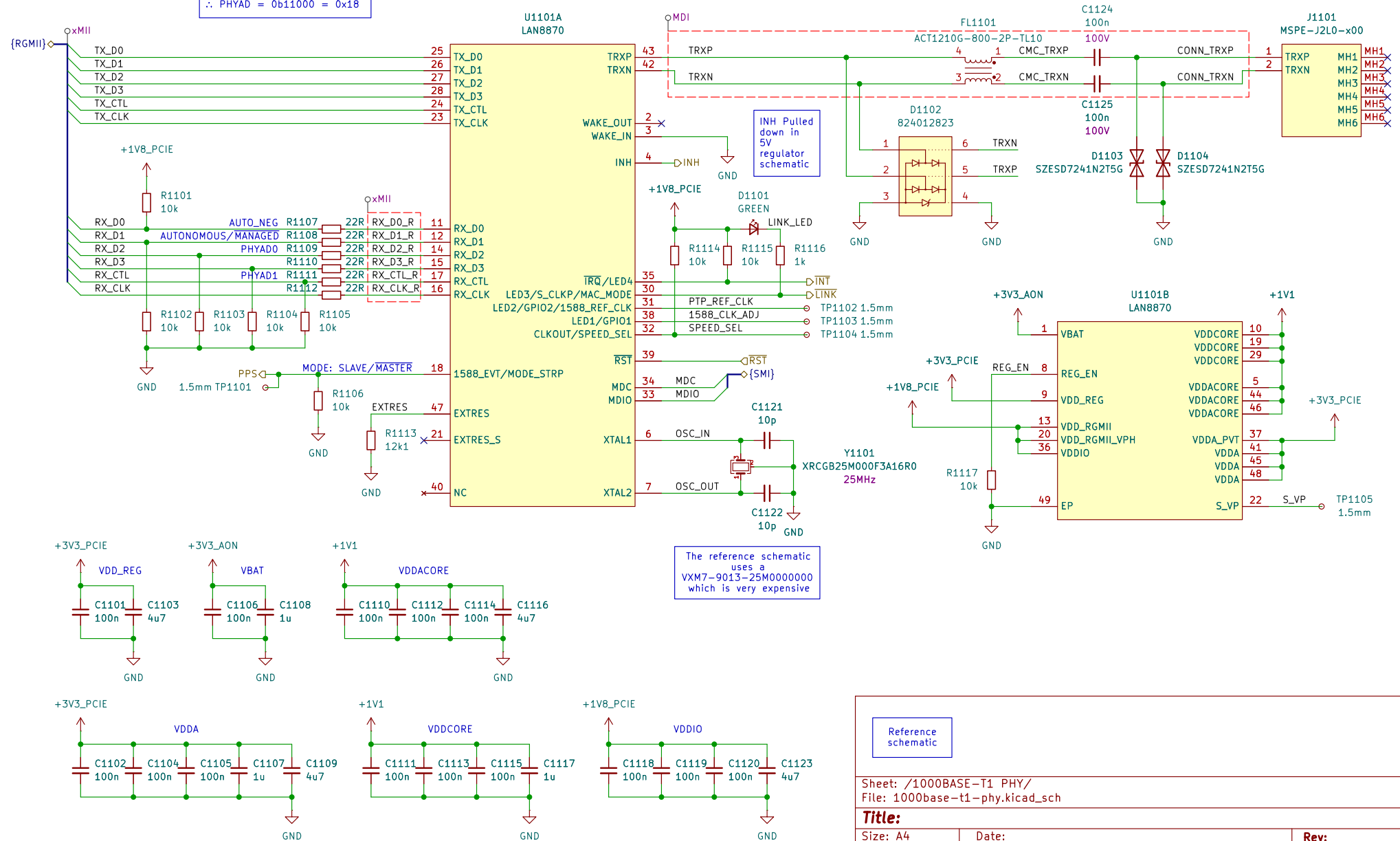
Rev:

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Single Pair Ethernet PHY & Connector

When in managed mode:
 - SPEED_SEL = PHYAD4
 - AUTO_NEG = PHYAD3
 - MODE = PHYAD2
 ∴ PHYAD = 0b11000 = 0x18



Reference schematic

Sheet: /1000BASE-T1 PHY/
 File: 1000base-t1-phy.kicad_sch

Title:

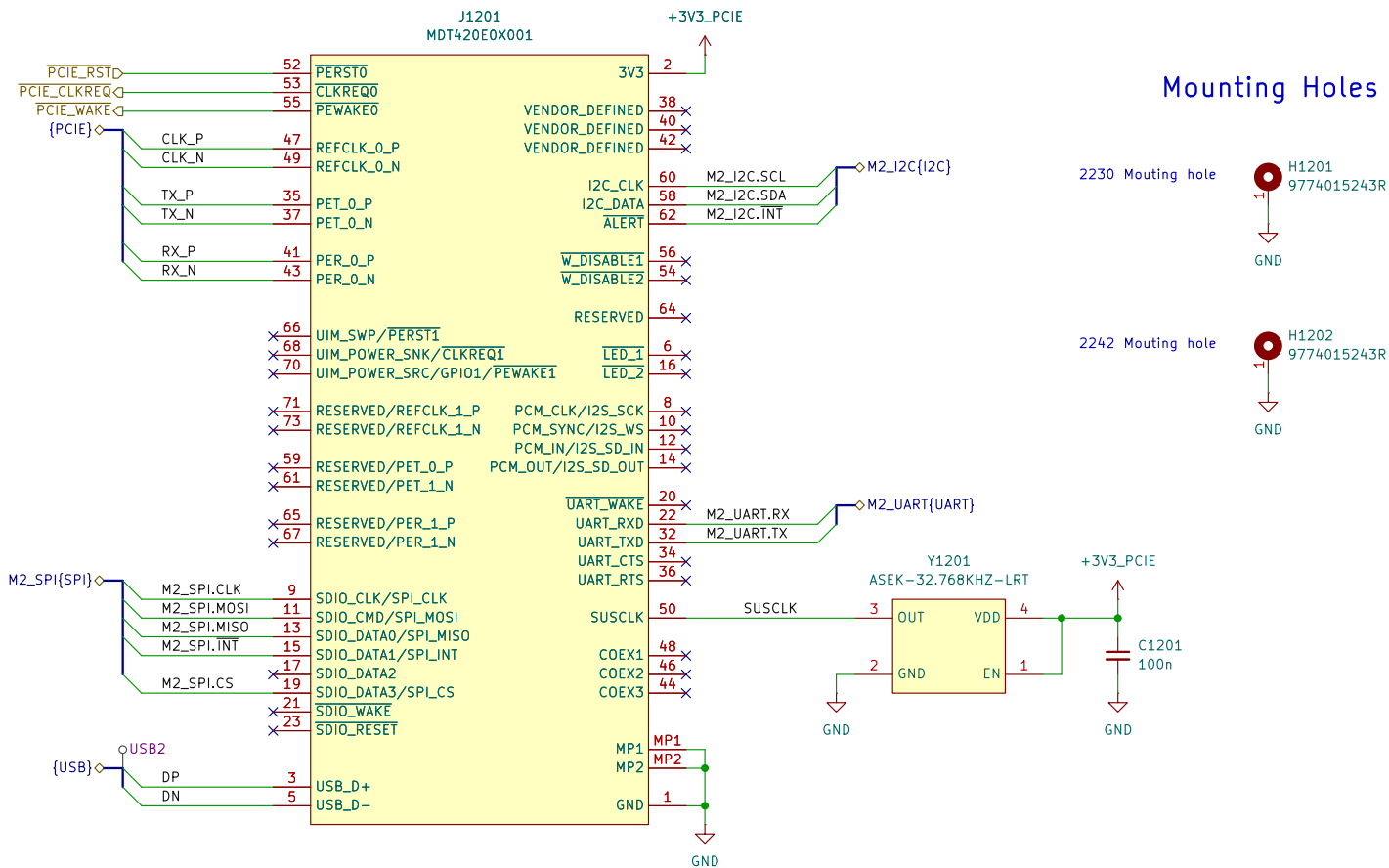
Size: A4 Date:

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M.2 E-Key



Sheet: /M.2 E-Key/
File: m-2-e-key.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.1

Date:

Rev:

Id: 12/12